

Definite integrals & FTC (Ch1)**Riemann sum / net area**

$$\int_a^b f(x) dx := \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x, \Delta x = \frac{b-a}{n}, x_i^* \in [x_{i-1}, x_i] \text{ (if limit exists, independent of sample points). Net area: } \int_a^b f = (\text{area above}) - (\text{area below}).$$

Core properties (integrable on $[a, b]$)

Linearity: $\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g$; $\int_a^a f = 0$; $\int_a^b f = -\int_b^a f$.

Additivity: $\int_a^b f = \int_a^c f + \int_c^b f$.

Order/Bounds: $f \geq g \Rightarrow \int_a^b f \geq \int_a^b g$; $m \leq f \leq M \Rightarrow m(b-a) \leq \int_a^b f \leq M(b-a)$.

Triangle ineq.: $\left| \int_a^b f \right| \leq \int_a^b |f|$; if $|f| \leq M$ then $\left| \int_a^b f \right| \leq M(b-a)$.

Symmetry: f odd $\Rightarrow \int_{-a}^a f = 0$; f even $\Rightarrow \int_{-a}^a f = 2 \int_0^a f$.

FTC (conditions matter)

FTC I: f cont on $[a, b]$, $F(x) = \int_a^x f(t) dt \Rightarrow F'(x) = f(x)$ on (a, b) .

Leibniz (moving limit): $\frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x))g'(x)$ (f cont, g diff).

More general: $\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f(v(x))v'(x) - f(u(x))u'(x)$.

FTC II: f cont on $[a, b]$, $F' = f \Rightarrow \int_a^b f(x) dx = F(b) - F(a)$.

Average value: $f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$ (cont).

MVT (integrals): f cont $\Rightarrow \exists c \in (a, b)$ s.t. $\int_a^b f = f(c)(b-a)$, so $f_{\text{avg}} = f(c)$.

 u -substitution

Indef: $u = g(x)$ diff $\Rightarrow \int f(g(x))g'(x) dx = \int f(u) du$.

Def: $g \in C^1[a, b]$, f cont on range(g): $\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$ (change limits).

Heuristic: spot inner $g(x)$ + missing $g'(x)$ (up to constant).

Improper integrals

Type 1 (infinite): $\int_a^\infty f := \lim_{t \rightarrow \infty} \int_a^t f$; $\int_{-\infty}^b f := \lim_{t \rightarrow -\infty} \int_t^b f$.

Type 2 (blow-up): if $f \rightarrow \infty$ at a^+ , then $\int_a^b f := \lim_{t \rightarrow a^+} \int_t^b f$ (split at interior singularities).

Convergence is required for the value to be defined; never cancel divergences across a split.

p -tests: $\int_1^\infty x^{-p} dx$ conv $\Leftrightarrow p > 1$; $\int_0^1 x^{-p} dx$ conv $\Leftrightarrow p < 1$.

Comparison: $0 \leq f \leq g$, $\int g$ conv $\Rightarrow \int f$ conv; $0 \leq g \leq f$, $\int g$ div $\Rightarrow \int f$ div.

Limit comparison: $f, g > 0$, $\lim_{x \rightarrow \cdot} \frac{f}{g} = c \in (0, \infty) \Rightarrow$ same conv/div.

Alternating/absolute: if $\int |f|$ conv $\Rightarrow \int f$ conv (absolute conv).

Techniques of integration (Ch3)**Integration by parts (IBP)**

IBP: $\int u dv = uv - \int v du$; Def IBP: $\int_a^b u dv = [uv]_a^b - \int_a^b v du$.

Choose u : Log, Inv trig, Algebraic, Trig, Exp (LIATE).

Tabular (poly $\times$ trig/exp): alternate signs; diff poly to 0; integrate other factor repeatedly.

Repeated-cycle: set $I = \int e^{ax} \sin bx dx$ or $\int e^{ax} \cos bx dx$, IBP twice $\Rightarrow$ solve linear equation for I .

Definite IBP pitfall: evaluate uv at endpoints *before* simplifying $\int v du$.

High-yield closed forms (from repeated IBP)

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C \quad (a, b \in \mathbb{R}).$$

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C.$$

Trig integrals: pattern selection (no identities here)

$\int \sin^m x \cos^n x dx$: n odd $\Rightarrow$ save $\cos x dx$, convert rest via *Formula Sheet*, $u = \sin x$; m odd $\Rightarrow$ save $\sin x dx$, $u = \cos x$; both even $\Rightarrow$ power-reduction (*Formula Sheet*).

$\int \tan^m x \sec^n x dx$: n even $\Rightarrow$ save $\sec^2 x dx$, $u = \tan x$; m odd $\Rightarrow$ save $\sec x \tan x dx$, $u = \sec x$.

Mixed arguments (e.g. $\sin kx \cos mx$): product-to-sum (*Formula Sheet*).

Trig substitution: radicals $\sqrt{\cdot}$

$$\sqrt{a^2 - x^2}: x = a \sin \theta, dx = a \cos \theta d\theta, \sqrt{a^2 - x^2} = a \cos \theta.$$

$$\sqrt{a^2 + x^2}: x = a \tan \theta, dx = a \sec^2 \theta d\theta, \sqrt{a^2 + x^2} = a \sec \theta.$$

$$\sqrt{x^2 - a^2}: x = a \sec \theta, dx = a \sec \theta \tan \theta d\theta, \sqrt{x^2 - a^2} = a \tan \theta.$$

Back-sub: triangle method or $\theta = \arcsin(x/a)$ etc (*Formula Sheet*). Complete square first for $\sqrt{ax^2 + bx + c}$.

Partial fractions (PF)

Pre-step: if $\deg P \geq \deg Q$ do long division. Factor Q over $\mathbb{R}$.

Templates: (i) $(ax + b)^r$: $\sum_{k=1}^r \frac{A_k}{(ax+b)^k}$; (ii) irreducible $(ax^2 + bx + c)^r$: $\sum_{k=1}^r \frac{A_k x + B_k}{(ax^2 + bx + c)^k}$.

Fast solve: distinct linear factors $\Rightarrow$ cover-up (at roots) + solve leftovers.

Quadratic: write $Ax + B = \lambda(2ax + b) + \mu \Rightarrow \lambda \ln |ax^2 + bx + c| +$ complete square for $\arctan(\cdot)$ (see *Formula Sheet*).

Applications of integration (Ch2)**Area between curves**

Vertical slices: $A = \int_{x=a}^b (\text{top} - \text{bottom}) dx$ (split where curves cross / switch order).

Horizontal slices: $A = \int_{y=c}^d (\text{right} - \text{left}) dy$ (often avoids splitting).

Bounds: solve intersections; ensure integrand ≥ 0 on each piece.

Volumes: general slicing

General: $V = \int A(x) dx$ (slices $\perp$ x -axis) or $V = \int A(y) dy$ (slices $\perp$ y -axis).

Revolution: build $A(\cdot)$ from radii; check units: $A (\text{length})^2 \Rightarrow V (\text{length})^3$.

Volumes: disks / washers (revolution)

About $y = c$ using dx : $R_{\text{out}}(x) = |y_{\text{top}}(x) - c|$, $R_{\text{in}}(x) = |y_{\text{bot}}(x) - c|$,

$$V = \pi \int_a^b (R_{\text{out}}^2 - R_{\text{in}}^2) dx.$$

About $x = c$ using dy : $R_{\text{out}}(y) = |x_{\text{right}}(y) - c|$, $R_{\text{in}}(y) = |x_{\text{left}}(y) - c|$,

$$V = \pi \int_c^d (R_{\text{out}}^2 - R_{\text{in}}^2) dy.$$

Rule: washers slice **perpendicular** to axis; shells slice **parallel**.

Volumes: cylindrical shells (revolution)

About $x = c$ using dx (vertical strips): $r(x) = |x - c|$, $h(x) = y_{\text{top}} - y_{\text{bot}}$, $V = 2\pi \int_a^b r(x)h(x) dx$.

About $y = c$ using dy (horizontal strips): $r(y) = |y - c|$, $h(y) = x_{\text{right}} - x_{\text{left}}$, $V = 2\pi \int_c^d r(y)h(y) dy$.

Arc length / surface area

If $y = f(x)$, $f' \in C[a, b]$: $L = \int_a^b \sqrt{1 + (f')^2} dx$; S about $y = c$:

$$S = 2\pi \int_a^b |f(x) - c| \sqrt{1 + (f')^2} dx.$$

If $x = g(y)$, $g' \in C[c, d]$: $L = \int_c^d \sqrt{1 + (g')^2} dy$; S about $x = c$:

$$S = 2\pi \int_c^d |g(y) - c| \sqrt{1 + (g')^2} dy.$$

(Parametric, if used): $L = \int_\alpha^\beta \sqrt{(x')^2 + (y')^2} dt$.

Average value

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx \text{ (cont on } [a, b]).$$

Work / pumping / hydrostatic force (templates)

Work (1D): $W = \int_a^b F(x) dx$ (force along line).

Hooke spring: $F = kx$ (displacement from natural length) $\Rightarrow W = \int_{x=a}^b kx dx$ (see *Formula Sheet*).

Pumping: slice at height y : $dW = \rho g A(y) D(y) dy$, so $W = \int dW$ (use weight density ρg).

Hydrostatic force: pressure $p(y) = \rho g h(y)$ (depth), $dF = p(y) w(y) dy$, $F = \int_{y=c}^d \rho g h(y) w(y) dy$.

Numerical integration (Ch3)**Rules**

Partition: $\Delta x = \frac{b-a}{n}$, $x_i = a + i\Delta x$.

Left: $L_n = \sum_{i=1}^n f(x_{i-1})\Delta x$; Right: $R_n = \sum_{i=1}^n f(x_i)\Delta x$.

Midpoint: $M_n = \sum_{i=1}^n f\left(\frac{x_{i-1} + x_i}{2}\right)\Delta x$.

Trapezoid: $T_n = \frac{\Delta x}{2} (f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n)) = \frac{L_n + R_n}{2}$.

Simpson (n even): $S_n = \frac{\Delta x}{3} (f(x_0) + 4 \sum_{\text{odd } i} f(x_i) + 2 \sum_{\text{even } i \neq 0, n} f(x_i) + f(x_n))$.

Error bounds (need smoothness)

$$\text{If } f'' \text{ cont on } [a, b], |f''| \leq K: |E_T| \leq \frac{K(b-a)^3}{12n^2}, |E_M| \leq \frac{K(b-a)^3}{24n^2}.$$

$$\text{If } f^{(4)} \text{ cont on } [a, b], |f^{(4)}| \leq K \text{ (} n \text{ even): } |E_S| \leq \frac{K(b-a)^5}{180n^4}.$$

Quick monotonicity check (common in tutorials): if $f'' \geq 0$ then T_n overestimates and M_n underestimates (reverse if $f'' \leq 0$).